```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import LinearRegression, ElasticNet
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
df = pd.read_csv('insurance.csv')

df.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.90	0	yes	southwest	16884.92
1	18	male	33.77	1	no	southeast	1725.55
2	28	male	33.00	3	no	southeast	4449.46
3	33	male	22.70	0	no	northwest	21984.47
4	32	male	28.88	0	no	northwest	3866.85

Next steps: Generate code with df New interactive sheet

```
le = LabelEncoder()

df['sex'] = le.fit_transform(df['sex'])
df['smoker'] = le.fit_transform(df['smoker'])
df['region'] = le.fit_transform(df['region'])

df.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	0	27.90	0	1	3	16884.92
1	18	1	33.77	1	0	2	1725.55
2	28	1	33.00	3	0	2	4449.46
3	33	1	22.70	0	0	1	21984.47
4	32	1	28.88	0	0	1	3866.85

Next steps: Generate code with df New interactive sheet

```
X = df[['age','bmi','children','smoker']]
y = df['charges']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
lr = LinearRegression()

lr.fit(X_train, y_train)

y_pred_lr = lr.predict(X_test)
```

```
lr_mae = mean_absolute_error(y_test, y_pred_lr)
lr_mse = mean_squared_error(y_test, y_pred_lr)
lr_r2 = r2_score(y_test, y_pred_lr)

print("Linear Regression Results")
print("MAE:", lr_mae)
print("MSE:", lr_mse)
print("R2:", lr_r2)
```

```
Linear Regression Results
MAE: 6951.764687745466
MSE: 69761426.4525848
R2: -0.6735273929874654
```

```
elastic = ElasticNet(alpha=0.5, l1_ratio=0.5)

elastic.fit(X_train, y_train)

y_pred_elastic = elastic.predict(X_test)
```

```
el_mae = mean_absolute_error(y_test, y_pred_elastic)
el_mse = mean_squared_error(y_test, y_pred_elastic)
el_r2 = r2_score(y_test, y_pred_elastic)

print("Elastic Net Results")
print("MAE:", el_mae)
print("MSE:", el_mse)
print("R2:", el_r2)
```

```
Elastic Net Results
MAE: 7028.457590749021
MSE: 55117321.085653394
R2: -0.3222256389441831
```

```
alphas = [0.1, 0.5, 1.0]
l1_ratios = [0.2, 0.5, 0.8]

results = []

for a in alphas:
    for l in l1_ratios:
        model = ElasticNet(alpha=a, l1_ratio=l)
```

```
        model.fit(X_train, y_train)
        pred = model.predict(X_test)

        mae = mean_absolute_error(y_test, pred)
        mse = mean_squared_error(y_test, pred)
        r2 = r2_score(y_test, pred)

        results.append([a, l, mae, mse, r2])

result_df = pd.DataFrame(results,
                         columns=['alpha','l1_ratio','MAE','MSE','R2'])

result_df
```

	alpha	l1_ratio	MAE	MSE	R2
0	0.1	0.2	6970.380975	6.285431e+07	-0.507831
1	0.1	0.5	6962.486993	6.510778e+07	-0.561890
2	0.1	0.8	6955.655214	6.774927e+07	-0.625257
3	0.5	0.2	7087.779541	5.246181e+07	-0.258522
4	0.5	0.5	7028.457591	5.511732e+07	-0.322226
5	0.5	0.8	6976.245048	6.153574e+07	-0.476199
6	1.0	0.2	7245.943965	5.277238e+07	-0.265972
7	1.0	0.5	7128.905708	5.185768e+07	-0.244029
8	1.0	0.8	7009.694947	5.669956e+07	-0.360182

Next steps: Generate code with `result_df` New interactive sheet

```
comparison = pd.DataFrame({
    "Model": ["Linear Regression", "Elastic Net"],
    "MAE": [lr_mae, el_mae],
    "MSE": [lr_mse, el_mse],
    "R2 Score": [lr_r2, el_r2]
})

comparison
```

	Model	MAE	MSE	R2 Score
0	Linear Regression	6951.764688	6.976143e+07	-0.673527
1	Elastic Net	7028.457591	5.511732e+07	-0.322226

Next steps: Generate code with `comparison` New interactive sheet